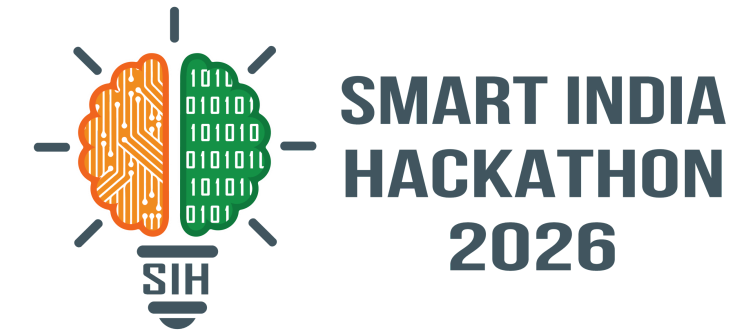
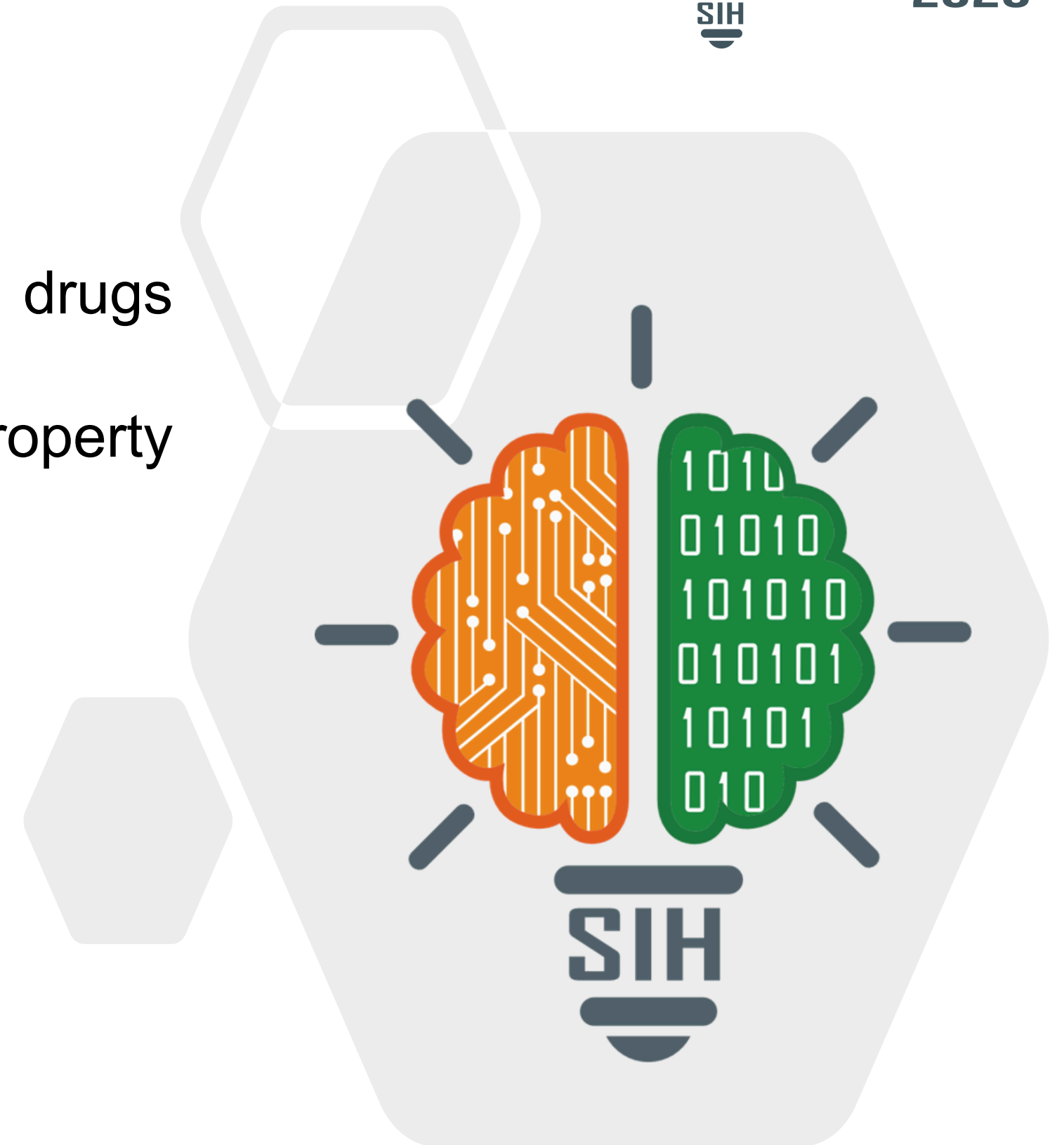


SMART INDIA HACKATHON 2026



- **Problem Statement ID** –SIH260166
- **Problem Statement Title-** A software that suggests drugs and formulations for a disease/pharmacological property based on the Ayurvedic classical books/Repositories
- **Theme**-MedTech/BioTech/ HealthTech
- **PS Category**- Software
- **Team ID-**
- **Team Name** - Tragic Bytes



Proposed Solution — Ayurveda Intelligence

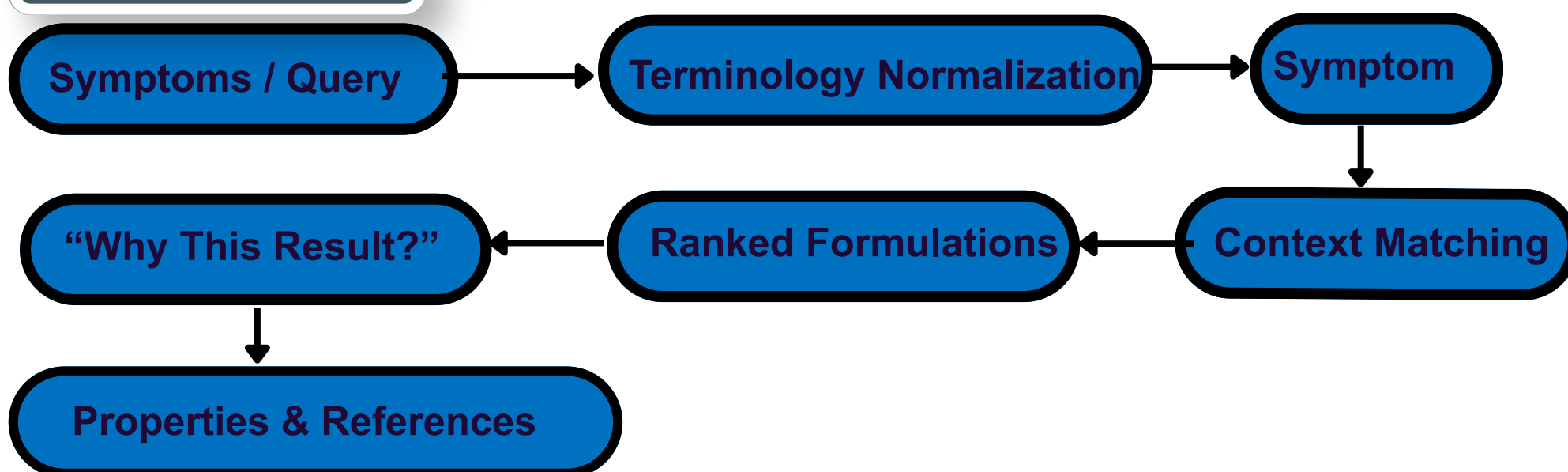
What We Are Building

- An evidence-aware Ayurvedic formulation discovery platform for students and practitioners.
- Identifies single drugs and compound formulations relevant to a user's symptoms, disease and clinical context.
- Organizes diseases, synonyms, formulations, ingredients, Ayurvedic properties and textual references into a structured knowledge system.

How It Addresses the Problem

- Reduces the need to manually search through 150+ scattered Ayurvedic texts and multiple terminology systems.
- Handles synonyms and context-dependent terminology instead of relying only on exact keyword matching.
- Finds formulations based on a combination of symptoms, helping distinguish between formulations with similar broad indications.
- Brings formulation details, properties, cautions and references together in one searchable interface.
- Makes results traceable and understandable by showing why a formulation was surfaced and where the information comes from.

How it works



Innovation & Uniqueness

- Symptom-Constellation Matching — matches multiple symptoms and clinical context, not just one disease keyword.
- Terminology Intelligence — handles synonyms, alternative names and ambiguous Ayurvedic terms.
- Context-Aware Interpretation — identifies the intended meaning of a term based on its context.
- Explainable Ranking — shows why a formulation was surfaced and how relevant it is.
- Evidence Traceability — links formulation results to available classical sources and references.
- Formulation Intelligence — combines ingredients, indications, Rasa, Guna, Virya, Vipaka and suitability information where available.

Existing approach: Search → Find Formulation → Read Details

Our approach: Symptoms + Context → Understand → Match → Rank → Explain → Verify

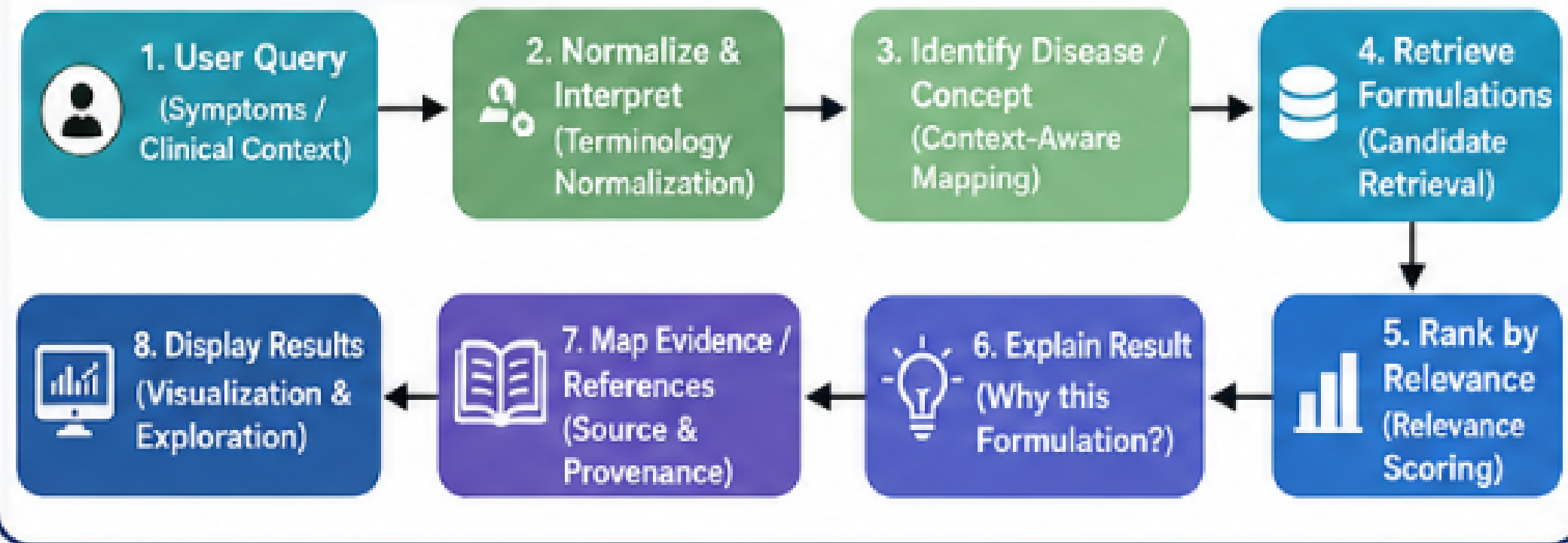
Technology & Architecture

- **Frontend:** React / Next.js — research interface & visualization
- **Backend:** Python + FastAPI — query processing & intelligence
- **Knowledge Layer:** Diseases, Formulations, Ingredients, Synonyms & Evidence
- **API Layer:** REST APIs + structured JSON communication
- **Data Layer:** Structured JSON / CSV / Database
- **Explainability Layer:** Match scores + “Why This Result?”
- **Evidence Layer:** Source & provenance tracking
- **Modular Architecture:** Ready for NLP, semantic search & knowledge graph expansion

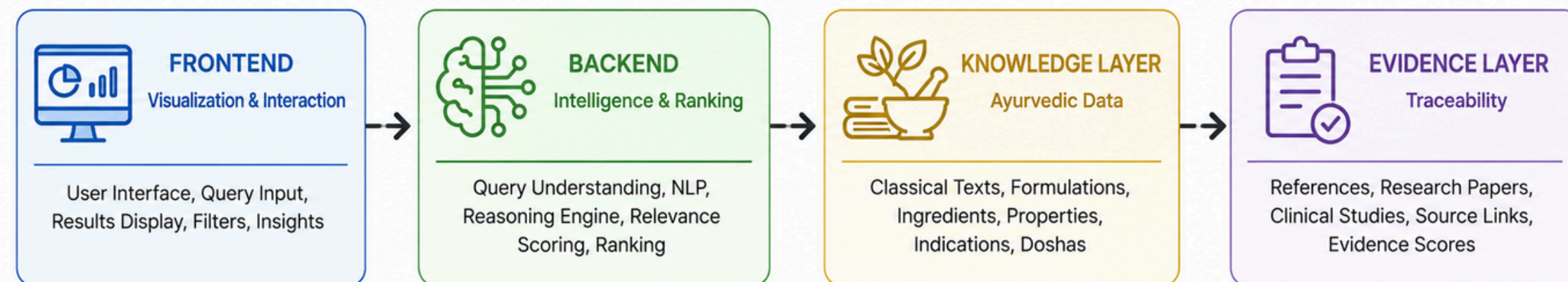
AI / Intelligence Layer

- **Terminology normalization**
- **Synonym resolution**
- **Context-aware interpretation**
- **Disease / concept identification**
- **Candidate formulation retrieval**
- **Relevance ranking**
- **Explainable scoring**
- **Future NLP / semantic search integration**

Implementation Flow



Technology Stack



Technical Feasibility

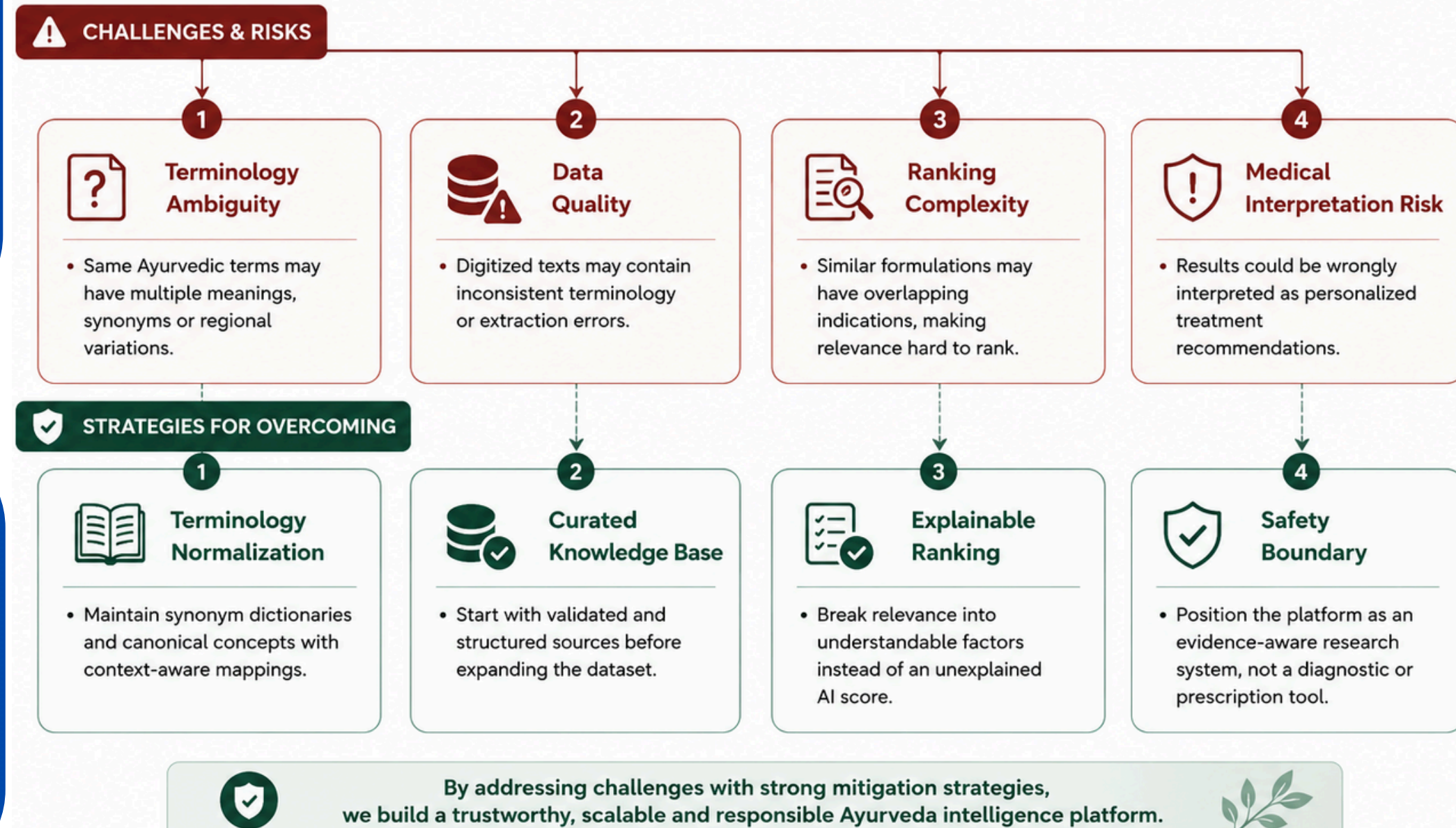
- Uses established technologies such as React / Next.js, Python, FastAPI and structured databases.
- Ayurvedic knowledge can be progressively converted into structured entities such as diseases, formulations, ingredients, synonyms and references.
- The modular architecture allows rule-based matching initially, with NLP and semantic search added later.
- REST APIs provide a clear connection between the frontend, backend, knowledge layer and future AI components.

Operational Feasibility

- Provides a centralized interface for exploring information currently distributed across multiple Ayurvedic texts and terminology systems.
- Supports students, researchers and practitioners without requiring them to manually navigate multiple sources.
- The explainability layer makes the system's reasoning easier to inspect and validate.
- Evidence and provenance tracking allows users to distinguish available source-backed information from unsupported results.

Challenges, Risks & Mitigation Strategies

Balancing challenges with effective mitigation for a reliable and responsible platform.



BENEFITS

IMPACT ON TARGET AUDIENCE

Students, Researchers & Practitioners

- Faster exploration of scattered Ayurvedic literature.
- Structured access to diseases, formulations, ingredients and terminology.
- Context-aware search across synonyms and alternative terms.
- Explainable formulation discovery with a visible research trail.
- Compare formulations using transparent relevance signals.

Social, Educational & Technological Impact

- Improves accessibility to structured Ayurvedic knowledge.
- Helps preserve traditional knowledge in a searchable digital format.
- Encourages evidence-aware learning and research.
- Reduces manual effort spent searching multiple sources.
- Creates a scalable foundation for NLP, semantic search and knowledge graphs



Social & Cultural



Enhances community knowledge and preserves traditions

Focuses on human connection and historical legacy



Improves access to structured Ayurvedic knowledge

Facilitates discovery for students and researchers



Supports responsible preservation for future generations

Ensures traditional wisdom survives in digital form



Economic & Environmental



Optimizes resource efficiency and digital sustainability

Focuses on operational productivity and ecological footprint



Reduces time spent searching multiple sources

Creates a scalable digital knowledge platform



Creates a long-term searchable digital repository

Reduces duplication of information and research resources

- [e-Charak — Ministry of AYUSH](#)

Medicinal plants, Ayurvedic drugs & traditional-medicine knowledge.

- [WHO — International Standard Terminologies on Ayurveda](#)

Standardized Ayurvedic terminology, concepts & synonyms.

- [CCRAS — Database on Medicinal Plants](#)

Research & classical-knowledge resources on medicinal plants.

- [GRAYU — Ayurvedic Knowledge Graph](#)

Graph-based integration of formulations, plants, phytochemicals & diseases.

PROJECT DOCUMENTATION

[Ayurveda Intelligence — GitHub Repository](#) · [System Architecture](#) · [Project Strategy](#) · [Project Report](#)